

Internship in Physical Organic Chemistry

Week 3, Part 3: Finding the most stable ion geometry

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Goals for this part:

- Generate and optimise the 3D geometry of your neutral compound
- Identify plausible protonation or deprotonation sites by chemical reasoning
- Optimise each candidate ion and compare their energies
- Select the most stable ion geometry as the reactant for the scan

1 Overview

In Week 2 you measured spectra in either positive or negative ion mode. The ion you fragmented in the mass spectrometer was either protonated ($[M+H]^+$, charge +1) or deprotonated ($[M-H]^-$, charge -1). Your computation must model the same ion — you cannot compute the neutral molecule and compare it to an ionic experiment.

The question is: **where is the proton added or removed?** A flavonoid has several oxygen-containing groups, each of which could accept or donate a proton. Each site gives a different ion with a different geometry and a different energy. To find the most likely structure, you will compute the energy of each candidate and select the lowest.

Before starting, open your terminal and run the startup commands.

Windows:

```
conda activate physorg
set PATH=C:\xtb\xtb-6.7.1\bin;%PATH%
```

macOS:

```
conda activate physorg
```

2 Generating the neutral geometry

Step 1 — Find the SMILES for your compound.

Go to <https://pubchem.ncbi.nlm.nih.gov> and search for your compound by name. On the compound page, scroll to the **Names and Identifiers** section and copy the **Canonical SMILES** string exactly as shown.

Step 2 — Generate a 3D geometry.

Move into your **neutral** folder and run OpenBabel with your SMILES:

Windows:

```
cd week3\neutral
obabel -:"PASTE_YOUR_SMILES_HERE" --gen3d -O neutral.xyz
```

macOS:

```
cd week3/neutral
obabel -:"PASTE_YOUR_SMILES_HERE" --gen3d -O neutral.xyz
```

Open `neutral.xyz` in your visualiser and verify the ring system is intact and the atom count matches the molecular formula of your compound.

Expected atom counts.

Luteolin (C₁₅H₁₀O₆): 31 atoms total.

Check the first line of your `neutral.xyz` file to confirm.

Step 3 — Optimise the neutral geometry.

```
xtb neutral.xyz --opt --chrg 0 --uhf 0 --gfn 2 --norestart > neutral_opt.out
```

Check for normal termination, then save a named copy of the optimised geometry before it is overwritten:

Windows:

```
type neutral_opt.out | findstr "normal termination"
copy xtbopt.xyz neutral_opt.xyz
```

macOS:

```
grep "normal termination" neutral_opt.out
cp xtbopt.xyz neutral_opt.xyz
```

Open `neutral_opt.xyz` in your visualiser. The geometry should look chemically reasonable with all three rings intact.

Record the SMILES string and the total energy of the neutral compound in section 3W-c of your `scratch.tex`:

Windows:

```
type neutral_opt.out | findstr "TOTAL ENERGY"
```

macOS:

```
grep "TOTAL ENERGY" neutral_opt.out
```

3 Identifying plausible ionisation sites

Before modifying the geometry on the computer, you must identify the candidate sites by chemical reasoning. Take a piece of paper and draw your compound. Mark all oxygen-containing groups clearly.

If you work in positive mode ($[M+H]^+$):

A proton adds to a site with available lone pair electrons. In a flavonoid, the candidate sites are the hydroxyl oxygens ($-OH$), the carbonyl oxygen ($C=O$), and the ring oxygen. Think about which lone pairs are most accessible and which protonated structure would be best stabilised — for example, by resonance or by forming a stable hydrogen bond.

Identify a few candidate sites. For each, draw the protonated structure and consider whether the resulting cation looks chemically stable.

If you work in negative mode ($[M-H]^-$):

A proton is removed from the most acidic $O-H$ bond. In a flavonoid, all $-OH$ groups are candidates. Think about which is most acidic — which $O-H$ bond is weakest, and which resulting anion is best stabilised by resonance or inductive effects. The position of the hydroxyl group relative to the aromatic rings and the carbonyl matters.

Identify a few candidate sites. For each, draw the deprotonated structure and consider whether the resulting anion is well stabilised.

Write your reasoning in section 3W-d of your `scratch.tex` — two or three sentences per candidate site explaining why it is plausible.

4 Modifying the geometry and optimising each ion

For each candidate site, you will: (1) create a subfolder, (2) modify the neutral geometry by adding or removing a proton, (3) optimise the ion, and (4) record the energy.

4.1 Creating a subfolder for each candidate

Windows:

```
cd week3\ion
mkdir site1
mkdir site2
mkdir site3
```

macOS:

```
cd week3/ion
mkdir site1
mkdir site2
mkdir site3
```

Use descriptive names instead of `site1`, `site2`, etc. — for example `carbonyl_0`, `C7_OH`, `ring_0`.

4.2 Modifying the geometry in the visualiser

Copy `neutral_opt.xyz` into each subfolder and modify it there.

Windows:

```
copy week3\neutral\neutral_opt.xyz week3\ion\carbonyl_0\start.xyz
```

macOS:

```
cp week3/neutral/neutral_opt.xyz week3/ion/carbonyl_0/start.xyz
```

Open `start.xyz` in your visualiser and make the modification:

Adding a proton (positive mode):

Chemcraft (Windows): go to **Edit** → **Add Atom**, select hydrogen, and click near the target oxygen. The hydrogen should be placed approximately 1 Å from the oxygen, at a chemically reasonable angle. Use the manipulation tool to adjust the position if needed. Save via **File** → **Save As** with the name `ion.xyz`.

Avogadro (macOS): select the Draw tool, choose hydrogen as the element, and click on the target oxygen to add a hydrogen. Adjust its position with the Manipulation tool. Save via **File** → **Export** → **Molecule** as `ion.xyz`.

Removing a proton (negative mode):

Chemcraft (Windows): click on the hydrogen you want to remove to select it (it will be highlighted). Press **Delete**. Save via **File** → **Save As** as `ion.xyz`.

Avogadro (macOS): select the hydrogen atom you want to remove by clicking on it. Press **Delete** or **Backspace**. Save via **File** → **Export** → **Molecule** as `ion.xyz`.

After saving, open `ion.xyz` in a text editor and check that the atom count changed by exactly one compared to the neutral. For protonation, the count increases by 1; for deprotonation, it decreases by 1. If the count is unchanged, the modification was not saved correctly.

4.3 Optimising the ion

Move into the subfolder and optimise. Use `--chrg 1` for a cation or `--chrg -1` for an anion:

Windows:

```
cd week3\ion\carbonyl_0
xtb ion.xyz --opt --chrg 1 --uhf 0 --gfn 2 --norestart > opt.out
copy xtbopt.xyz ion_opt.xyz
```

macOS:

```
cd week3/ion/carbonyl_0
xtb ion.xyz --opt --chrg 1 --uhf 0 --gfn 2 --norestart > opt.out
cp xtbopt.xyz ion_opt.xyz
```

Check for normal termination and extract the total energy:

Windows:

```
type opt.out | findstr "TOTAL ENERGY"
```

macOS:

```
grep "TOTAL ENERGY" opt.out
```

Repeat for all candidate sites, each in its own subfolder.

5 Comparing energies and selecting the reactant

You now have a total energy in Hartree for each candidate ion. Because all ions have the same number of atoms and the same charge, their energies can be directly compared — the lowest energy is the most stable structure.

Collect all energies in section 3W-e of your `scratch.tex`. Convert to relative energies in kcal/mol by subtracting the lowest energy from each value and multiplying by 627.5:

$$\Delta E_{\text{rel}} = (E - E_{\text{min}}) \times 627.5 \quad [\text{kcal/mol}]$$

Your table should look like this (values are illustrative):

Site	Total energy (Eh)	Rel. energy (kcal/mol)
cite XXX	-61.8234...	0.0 <- most stable
cite XXX	-61.8197...	+2.3
cite XXX	-61.8181...	+3.3
cite XXX	XXXXX	XXXXX
cite XXX	XXXX	XXXX
....		

The ion with the lowest total energy (relative energy = 0.0) is the most stable. Copy its optimised geometry into your `scan` folder:

Windows:

```
copy week3\ion\carbonyl_0\ion_opt.xyz week3\scan\reactant.xyz
```

macOS:

```
cp week3/ion/carbonyl_0/ion_opt.xyz week3/scan/reactant.xyz
```

Does the most stable ion match your MS data?

Look at your Week 2 spectra. The fragmentation pattern you observed depends on the structure of the precursor ion. Do the fragments you identified correspond to the ion you have selected as the most stable? If the most stable protonation site and the fragmentation pattern are inconsistent, discuss this with your supervisor — it may indicate that a less stable but more reactive ion is responsible for the fragmentation.

Record in section 3W-e:

- the energy table for all candidate sites
- which site you selected and why
- whether the selection is consistent with your Week 2 fragmentation pattern

You are ready to proceed to Part 4, where you will set up and run the potential energy scan on this reactant geometry.